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A SOVEREIGNTY PROPOSAL FOR THE KOREA–CANADA–ALLIES DECADE

Sovereign Compute *North.*

An energy-for-memory swap. A government-owned data substrate. A Canadian sovereignty standard built in partnership with Korea — and exported to allies who are not the United States.

The proposition. Canada has the cheapest low-carbon firm electricity in the G7. Korea controls the global memory chokepoint on which every modern AI accelerator depends. The United States has stated, under oath through Microsoft France in June 2025, that it cannot guarantee allied citizens' data against US extraterritorial reach. The trade is obvious. Canadian electrons, Korean memory, Canadian law, Korean and allied capital — combined into a government-owned compute fabric that reaches from the Hydro-Québec switchyard to the municipal town hall.

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CO-ISSUED WITH FIT FOR GOV

FOR CANADIAN & ALLIED USE

§ I THE THESIS · ONE PAGE

An energy-for-memory swap, signed under Canadian law.

Every modern AI accelerator that ships from Santa Clara contains memory stacked by a Korean fab. Every Korean fab is fed by Korean electricity its national utility cannot keep affordable. Every Canadian province between the Manitoba border and the Atlantic operates a surplus of firm low-carbon power it currently exports to United States wholesale markets at prices that subsidise American load. These three facts have not been combined into a Canadian industrial strategy. They should be.

The Carney government has already opened the door. On 11 September 2025 in Edmonton, the Prime Minister directed the Major Projects Office to develop a Canadian Sovereign Cloud. On 13 November 2025 in Terrace, he named data sovereignty one of four nation-building strategies, alongside energy, critical minerals and trade corridors. On 30 October 2025 in Gyeongju, he signed the Canada–Korea Security and Defence Cooperation Partnership — the first of its kind for Canada in the Indo-Pacific.[1] What has not yet been done is to fuse these decisions into a single, financeable architecture.

This working paper proposes that architecture. The shorthand is **Sovereign Compute North**: a government-owned, allies-financed, Canadian-law-bound compute fabric, anchored on surplus Quebec, Manitoba, BC and Ontario power, capitalised through a Korea-led consortium of Asian sovereign capital, governed by a new Canadian Data Sovereignty Standard, and extended down to every Canadian municipality that needs a substrate that does not export its citizens' data to a foreign court order.

~950 TWh
IEA PROJECTED GLOBAL DATA-CENTRE DEMAND BY 2030[2]

~75–80%
COMBINED KOREAN (SKH + SAMSUNG) SHARE OF HBM[3]

CAD $0.036
HYDRO-QUÉBEC FRAMEWORK RATE, FLEXIBLE LONG-TERM LOADS, PER KWH[4]

"Non, je ne peux pas"
MICROSOFT FRANCE, FRENCH SENATE, SWORN, 10 JUNE 2025[5]

WHAT THIS PAPER IS AND IS NOT

This is not a plea for industrial subsidy, a campaign for a new federal department, or a complaint about American hegemony. It is a Canadian sovereignty proposal that happens to be financeable, that happens to align with what the current federal government has already announced, and that happens to make Canada the highest-utility partner Korea could choose for compute-side industrial diversification outside the United States. It is also, on the municipal layer, the only architecture that closes the gap between what Buy Canadian binds and what the CLOUD Act reaches.

1. Office of the Prime Minister of Canada, "Prime Minister Carney announces the first projects to be reviewed by the new Major Projects Office," Edmonton, 11 September 2025; PM Carney, second-tranche announcement, Terrace BC, 13 November 2025; Joint Statement by PM Carney and President Lee Jae Myung, Gyeongju, 30 October 2025.
2. International Energy Agency, Key Questions on Energy and AI, 16 April 2026; Energy and AI, 10 April 2025.
3. Counterpoint Research, HBM market-share estimates, Q2–Q3 2025; TrendForce / Yonhap, 28 January 2026; Digitimes, 20 August 2025.
4. Hydro-Québec framework offering to qualifying flexible long-term industrial loads; rate quoted in CAD per kWh.
5. Anton Carniaux, Microsoft France Director of Public and Legal Affairs, sworn testimony before the French Senate Commission of Inquiry on digital sovereignty, 10 June 2025.

§ II THE COMPUTE BOTTLENECK IS ELECTRICITY, NOT SILICON

The world is being rationed by megawatts. Canada has megawatts.

For the first time in modern industrial history, the binding constraint on a frontier industry is not capital, not talent, not intellectual property, and not even chips. It is the electron itself.

The International Energy Agency's Energy and AI report (10 April 2025) projected global data-centre electricity demand to more than double from 415 TWh in 2024 to roughly 945 TWh by 2030 — about three percent of total global electricity. The April 2026 update reset the base to 485 TWh and held the 2030 projection near 950 TWh, with the AI-specific subset "poised to triple."[1] In the United States, data centres are on track to account for nearly half of all electricity demand growth this decade. In Japan, more than half. IEA Executive Director Fatih Birol:

> “ Global electricity demand from data centres is set to more than double over the next five years, consuming as much electricity by 2030 as the whole of Japan does today.
>
> — Fatih Birol, Executive Director, International Energy Agency, 10 April 2025

The price signal is now visible. The conditional offtake pipeline between data-centre operators and small modular reactor projects "has grown from 25 gigawatts at the end of 2024 to 45 gigawatts" by April 2026.[2] The Stargate project (OpenAI, SoftBank, Oracle, MGX) has publicly committed US$500 billion to 10 GW of US-domiciled AI infrastructure, with approximately 7 GW and over $400 billion announced as of late September 2025.[3] A frontier training run — GPT-4-class, approximately 25 megawatts continuous over ninety days — consumes on the order of 50 GWh of electricity, costs over US$100 million in capex-and-opex combined, and (in MIT Technology Review's words) "could power San Francisco for three days."[4]

WHY THE UNITED STATES IS THE BINDING CONSTRAINT

American grid expansion is not keeping pace with American AI capex. New interconnections in PJM, ERCOT and SPP now run two-to-five years behind hyperscaler requirements. The Stargate joint venture itself — Bloomberg reporting (August 2025) and The Information reporting (February 2026) — had hired no staff and was renegotiating its three-way structure six months after announcement; original timelines slipped; Oracle and OpenAI moved toward a two-way deal.[5] The capital exists. The compute design exists. The electrons do not.

WHY KOREA CANNOT SOLVE THIS AT HOME

Korea Electric Power Corporation carried approximately 206 trillion won (US$139 billion) in accumulated debt as of mid-2025. Industrial tariffs were raised 9.7 percent in October 2024 to roughly 182.7 won per kilowatt-hour for the largest users. Rates were frozen in Q4 2025 and Q1 2026 on government instruction — KEPCO would otherwise have been forced to cut to negative 13 won per kilowatt-hour given falling fuel costs.[6] Korean business power runs near KRW 173.982 per kilowatt-hour (~USD 13.2¢) against Hydro-Québec's framework rate for qualifying flexible loads near CAD $0.036 (~USD 2.6¢). Korean industrial expansion of compute is therefore a fiscal impossibility for the utility and a fiduciary problem for the chaebol.

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1. IEA, Energy and AI, 10 April 2025; IEA, Key Questions on Energy and AI, 16 April 2026.
2. IEA, Key Questions on Energy and AI, executive summary, April 2026.
3. OpenAI, "Announcing The Stargate Project," 21 January 2025; OpenAI, "Five new Stargate sites," 23 September 2025.

4. MIT Technology Review, "We did the math on AI's energy footprint," 20 May 2025; Towards Data Science, "The carbon footprint of GPT-4," 2024.
5. Bloomberg, August 2025; The Information, February 2026; The Decoder, "Stargate's $500 billion AI infrastructure project reportedly stalls," 2025.
6. Korea Herald, "KEPCO keeps electricity rate frozen for Q1," December 2025; S&P Global, "South Korea raises electricity rates for industrial use," October 2024; GlobalPetrolPrices, September 2025.

§ III THE KOREAN MEMORY CHOKEPOINT · AND WHY KOREA WANTS OUT

Korea controls the memory chokepoint and knows it is fragile.

Every modern AI accelerator that matters — Nvidia H100, H200, B200, GB200, B300, and the forthcoming Vera Rubin platform — depends on stacked high-bandwidth memory manufactured almost entirely by two Korean firms. The numbers are not in dispute.

A · THE HBM AND DRAM POSITION

MEMORY CLASS · Q3 2025	SK HYNIX	SAMSUNG	MICRON / OTHER
HBM (bit-shipment / revenue blend)	~53–62%	~17–35%	~11–21%
DRAM (revenue)	34%	33%	26% + CXMT 5% + Nanya 2%
Nvidia HBM3E supply share (mid-2025)	~90%	residual	residual
Nvidia HBM4 / Vera Rubin orders (early 2026)	~70%	early delivery target	balance

Counterpoint Research's measures of Korean HBM share blend bit-shipment and revenue methodologies; the combined Korean position consistently lands between 75 and 80 percent of the global market. HBM capacity is sold out across all three suppliers through 2026. Micron projects the HBM total-addressable market rising from approximately US$35 billion (2025) toward US$100 billion by 2028.[1]

B · THE STRUCTURAL VULNERABILITY KOREA HAS ADMITTED

Korea's semiconductor sector is now so concentrated that it has become a national-economic risk in its own right. The Diplomat in November 2025: South Korea's "semiconductor dependence is becoming a structural economic risk."[2] The Lee

administration has responded with the largest semiconductor industrial policy in Korean history — the K-Semiconductor Strategy targeting roughly US$450 billion in private investment and US$260 billion in tax incentives, the K-Semiconductor Belt cluster, the Yongin mega-cluster, the K-Cloud Strategy and the 4th Basic Plan for Cloud Computing (2025–2027).[3] On 10 December 2025 President Lee chaired "K-Semiconductor Vision and Promotion Strategy in the AI Era," committing 4.5 trillion won in public-private investment toward a mutual-growth foundry and a defence-semiconductor self-reliance project.[4]

C · WHY KOREA BENEFITS FROM A CANADIAN COMPUTE PARTNER

Three reasons, in increasing weight. First, Korean ESG-conscious sovereign capital — National Pension Service, Korea Investment Corporation, Korea Investment PE — is now bound by global disclosure standards that make Canadian hydro-powered compute the highest-rated jurisdiction available for AI training emissions. Second, every gigawatt of Korean memory whose downstream offtake is captured inside the United States is a gigawatt of Korean exposure to American export controls, tariff drift and political volatility — diversifying Korean offtake to a jurisdiction with a Canada-Korea Free Trade Agreement (in force since 2015) and a Comprehensive Strategic Partnership (2022) reduces that exposure. Third, the United States already has Stargate. Canada offers Korea what no American site can: equity in a sovereign vehicle, anchor offtake in a CLOUD-Act-free jurisdiction, and a defence-industrial track that runs from submarines to memory to data centres on one continuous strategic line.

> “ Our greatest strength has, in a structural sense, become our greatest fragility.
>
> — Paraphrased thesis from The Diplomat, November 2025, on Korean semiconductor concentration

1. Counterpoint Research, HBM market-share estimates, Q2–Q3 2025; Digitimes, "Nvidia's HBM supply chain to undergo major reshuffle in 2026," 20 August 2025; TrendForce reporting Yonhap, "SK hynix reportedly to supply about two-thirds of Nvidia HBM4," 28 January 2026; Micron HBM TAM projection, 2025.
2. Robert Kelly et al., "South Korea's Semiconductor Dependence Is Becoming a Structural Economic Risk," The Diplomat, November 2025.
3. Korea.net, K-Semiconductor Strategy briefings and updates, 2024–2025; ACM Communications, "South Korea's Nationwide Effort for AI Semiconductor Industry," 2025.
4. Office of the President of the Republic of Korea, "K-Semiconductor Vision and Promotion Strategy in the AI Era," 10 December 2025.

§ IV CANADIAN ENERGY ABUNDANCE · PROVINCE BY PROVINCE

The only G7 country with the electrons and the trust profile.

Canada installed approximately 158 gigawatts of generation capacity in 2025, with hydropower at 55 to 60 percent of generation, nuclear at roughly 13.5 percent, and 78 percent of all generation non-emitting. Canada is the world's third-largest hydro producer and second-largest uranium producer.[1] The provincial picture is what matters.

PROVINCE · UTILITY	INSTALLED / SURPLUS	INDUSTRIAL / DC RATE	CARBON INTENSITY
Quebec · Hydro-Québec	~38 GW · 99% renewable	CAD $0.036/kWh (framework) CAD $0.13/kWh (filed 18 Feb 2026)	~30 g CO_2/kWh
Ontario · IESO / OPG / Bruce	12,844 MW nuclear +4,800 MW Bruce C pre-development +1,200 MW Darlington SMR	~CAD 11–13¢/kWh	~40 g CO_2/kWh
BC · BC Hydro	+1,100 MW Site C (FY2026) ~8,000 GWh/yr export surplus Schedule 1894 DC discount	CAD ~7¢/kWh (industrial)	~15 g CO_2/kWh
Manitoba · Manitoba Hydro	~8,000 GWh/yr surplus currently exported to US	Among lowest in N. America	~5 g CO_2/kWh
Alberta · ENMAX / TransAlta	23,164 MW · coal phased out Jun 2024 26% gas cogen · 25% wind	Market-based	~300+ g CO_2/kWh

A · HYDRO-QUÉBEC, THE ANCHOR

On 18 February 2026 Hydro-Québec filed a formal new data-centre tariff at CAD 13 cents per kilowatt-hour for facilities above 5 MW, with a separate 19.5-cent rate for blockchain operations and a five-year transition period for existing large-power customers.[2] Hydro-Québec projects sevenfold data-centre growth in the province — from roughly 200 MW

today to approximately 1,000 MW by 2035. The framework offering to flexible long-term industrial loads sits near CAD $0.036/kWh, materially below Korean industrial rates and roughly one-third of US PJM industrial. President-CEO Claudine Bouchard:

> “ These rate adjustments will promote responsible electricity use and ensure we fully realize the value of our energy, taking into account prices elsewhere in North America.
>
> — Claudine Bouchard, President-CEO, Hydro-Québec, 18 February 2026

B · ONTARIO'S NUCLEAR POSITION

Ontario operates 12,844 MW of installed nuclear capacity across Bruce, Darlington and Pickering. Bruce Unit 6 returned from refurbishment in September 2023. The Bruce C impact-assessment notice of commencement issued in August 2025 covers up to 4,800 MW of new nuclear. The Darlington SMR programme targets 1,200 MW by 2036, with the BWRX-300 lead unit under construction. The Pickering B refurbishment received approval in January 2024, and OPG initiated environmental review at Wesleyville / Port Hope in January 2026 for up to 10,000 MW of additional nuclear capacity.[3]

C · BC AND MANITOBA — THE STRANDED SURPLUS

Site C entered service in fiscal 2026 at 1,100 MW and 5.1 TWh per year. BC Hydro's Schedule 1894 — the Clean Industry and Innovation Rate — offers data-centre and clean-fuel customers a transmission-service discount through March 31, 2030, with a 1,500 GWh per year allocation. Manitoba Hydro operates a Surplus Energy Program and exports approximately 8,000 GWh per year to United States wholesale markets at depressed prices — a stranded asset by any reasonable accounting, since Manitoba Hydro itself states publicly that export revenue keeps Manitoban rates roughly 20 percent lower than they would otherwise be. Manitoban electrons are currently subsidising American load. They should be powering Canadian compute instead.[4]

THE STRUCTURAL FACT

Canada is the only country in the G7 that combines surplus low-carbon firm electricity, a Korea-trusted political risk profile, and a federal mandate to build sovereign

infrastructure. The window in which this combination remains exclusive is short.

1. Natural Resources Canada, Energy Fact Book, Spring 2026 Edition; US International Trade Administration, Canada Country Commercial Guide — Energy, 2025.
2. Hydro-Québec press release, 18 February 2026; Data Center Dynamics, "Canadian utility Hydro-Québec proposes electricity tariff for data centers."
3. World Nuclear Association, Canada profile, 2025–2026; Canada Energy Regulator, Ontario provincial profile.
4. Manitoba Hydro, "Operations and Exports" (corporate page); BC Hydro, "Industrial Electrification Rates" (Schedule 1894); Statistics Canada, Investment in Non-Emitting Electricity Generation, 2025.

§ V THE CLOUD ACT BASELINE · THE WALL SOVEREIGN CLOUD BOUNCES OFF

The vendors have now admitted it under oath. Treat the admission as load-bearing.

On 10 June 2025, before the French Senate's Commission of Inquiry on digital sovereignty, Microsoft France's Director of Public and Legal Affairs Anton Carniaux was asked, under oath, whether he could guarantee that French citizens' data hosted on Microsoft cloud infrastructure in France would not be transferred to United States authorities without French consent. The transcript records his exact answer:

> “ Non, je ne peux pas le garantir.
>
> — Anton Carniaux, Microsoft France, Senate Commission of Inquiry, 10 June 2025

Carniaux added that Microsoft is contractually committed to resist requests it considers unfounded but cannot block lawful, specific orders issued under the United States Clarifying Lawful Overseas Use of Data Act (CLOUD Act, 18 U.S.C. § 2713, in force 2018). Microsoft France's Technical Director Pierre Lagarde confirmed the technical EU Data Boundary commitments but did not contest the legal point. Amazon Web Services published a "Five Facts" response in July 2025 framing the CLOUD Act as narrow, but did not contest its extraterritorial reach over any data held by a US-parent entity regardless of physical location. [1]

A · WHAT THIS MEANS IN PLAIN LANGUAGE

"EU Data Boundary," "Canadian region," and "in-country residency" are engineering features that do not override the CLOUD Act's extraterritorial reach so long as the operating entity has a United States parent. Microsoft Azure Canada Central in Toronto, AWS Canada Central in Montréal, AWS Canada West in Calgary, and Google Cloud's Canadian regions are all physically in Canada and legally in the United States. The Carniaux testimony makes this no longer a contestable assertion — it is admitted vendor doctrine.

B · WHAT EVERY CANADIAN PEER HAS ALREADY DONE THAT CANADA HAS NOT

France	Bleu, Numspot, S3NS · Gaia-X · EUCS "high+" tier
Germany	STACKIT (Schwarz Group) · Delos · BSI C5 sovereign tier
Australia	ASD-administered "Certified Sovereign" tier · Hosting Certification Framework
European Union	EUCS Cybersecurity Certification Scheme for Cloud Services · "high+" tier
Republic of Korea	4th Basic Plan for Cloud Computing 2025–2027 · K-Cloud · K-Cloud security framework
Canada · status quo	Government of Canada Cloud Adoption Strategy + ITSG-33 + Cloud Guardrails No vendor-domicile rule. No CLOUD Act exclusion.

C · THE BUY CANADIAN GAP

The federal Buy Canadian procurement framework, in force since 16 December 2025 for federal contracts of $25 million and above, binds origin of goods and services. It does not bind jurisdiction of the controlling parent entity. A "Canadian region" of a United States hyperscaler, operated by a Canadian subsidiary with Canadian staff, satisfies Buy Canadian and fails the CLOUD Act test. The Canadian Data Sovereignty Standard proposed in section VIII of this paper closes that gap. Nothing else now in force does.

1. French Senate Commission of Inquiry on digital sovereignty, transcript of 10 June 2025; The Register, "Microsoft exec admits it 'cannot guarantee' data sovereignty," 25 July 2025; AWS, "Five Facts on the CLOUD Act," July 2025; CONVOTIS legal advisory, "Microsoft: U.S. Access to EU Data Confirmed," 2025.

§ VI THE KOREA–CANADA TRACK · WHAT IS ALREADY SIGNED

This is not a proposal to start a relationship. It is a proposal to extend one.

The Canada–Korea bilateral relationship is the most operationally advanced non-US defence-industrial track Canada has. The instruments are already in place. The question is whether Canada chooses to extend them to include a sovereign compute fabric.

Canada-Korea Free Trade Agreement	In force 1 Jan 2015 · 99% of Canadian exports duty-free
Comprehensive Strategic Partnership	Concluded 23 Sep 2022 · Action Plan published July 2024
2+2 Foreign & Defence Ministerial	Inaugural meeting November 2024 · a treaty-ally-tier mechanism
Defence Materiel Cooperation MOU	Renewed December 2022
Security and Defence Cooperation Partnership	Signed 30 October 2025 · Gyeongju · "first of its kind for Canada in the Indo-Pacific"
Classified Information Agreement	Signed February 2026 · Anand & Cho
Hanwha Ocean · Algoma Steel MOU	26 January 2026 · CAD $345 M · Sault Ste. Marie structural-steel mill
KOGAS · LNG Canada	5% stake retained · 700,000 tpy offtake from 2024
ROKS Dosan Ahn Chang-ho	CFB Esquimalt visit 23–24 May 2026 · joint ASW exercises with RCN

A · THE SDCP FRAMING — IN THE LEADERS' OWN WORDS

The 30 October 2025 joint statement from Prime Minister Carney and President Lee Jae Myung describes the Security and Defence Cooperation Partnership as "the first of its kind for Canada in the Indo-Pacific" and as providing "a clear, action-oriented framework for accelerating defence cooperation, enhancing interoperability and readiness, strengthening defence industrial cooperation, and jointly contributing to regional and global stability." The supply-chain scope explicitly includes "energy, critical minerals and other materials … essential for defence production." It establishes a formalised Space Security Dialogue and AI / autonomy cooperation in the military domain.[1]

B · WHAT HANWHA HAS ALREADY DONE

The Hanwha Ocean — Algoma Steel binding MOU of 26 January 2026 commits up to CAD $345 million (US$250 million), contingent on Hanwha winning the Canadian Patrol Submarine Project: $200 million to back a new structural-steel beam mill in Sault Ste. Marie,

plus $50 million in product purchases. A Hanwha-commissioned April 2025 study projects 15,000 to 22,500 jobs annually through 2044 across Hanwha's Canadian investments. Hanwha has parallel MOUs with Telesat, MDA Space, Cohere and PV Labs for satellite, AI and advanced-tech collaboration announced in January 2026.[2]

C · WHAT THIS PAPER PROPOSES ADDING TO THAT TRACK

A Korea-led consortium — Samsung, SK Hynix, Hanwha, plus sovereign LPs (National Pension Service, Korea Investment Corporation, Korea Investment PE) — as a strategic minority alongside a Canadian Crown-corporation majority, capitalising Sovereign Compute North on Canadian electrons under the Canadian Data Sovereignty Standard. The structural logic is symmetric: Korean memory wealth flows to Canadian electron wealth, the joint venture is owned by Canadian Crown majority and governed under Canadian law, and the offtake is reserved for Canadian and allied sovereign workloads first, commercial second.

> “ The first of its kind for Canada in the Indo-Pacific.
>
> — Joint Statement, PM Carney & President Lee Jae Myung, Gyeongju, 30 October 2025

1. Office of the Prime Minister of Canada, "Joint statement by Prime Minister Mark Carney of Canada and President Lee Jae Myung of the Republic of Korea," 30 October 2025; Global Affairs Canada, Canada-Republic of Korea Security and Defence Cooperation Partnership announcement, 29 October 2025.
2. BNN Bloomberg, "Korean shipbuilder bidding for submarine contract agrees to $345M deal with Algoma Steel," 26 January 2026; CBC News, Sault Ste. Marie News and Canadian Metalworking reporting, January 2026; Canada.ca, "Canada and Republic of Korea strengthen ties in key industrial sectors," January 2026.

§ VII THE ARCHITECTURE · GOVERNMENT-OWNED, ALLIES-FINANCED, MUNICIPALLY EXTENDED

Crown majority. Korean minority. Canadian law. Municipal floor.

The proposed structure is straightforward, financeable today, and politically defensible because every one of its components is already in motion under federal policy.

A · CAPITAL STACK

Canadian Crown majority	51%+ · MPO-anchored vehicle · CDPQ, CPPIB, BCI, Alberta Heritage co-LPs

Korean strategic minority	≤30% · Samsung, SK Hynix, Hanwha + NPS / KIC / Korea Investment PE
Allied minority	≤15% · Japan (GPIF), Australia (Future Fund), UK, EU members under EUCS-aligned terms
Federal anchor	CAD $2B Sovereign AI Compute Strategy (Dec 2024) + Building Canada Act facility

B · PHYSICAL ARCHITECTURE

Anchor site (Tier 1)	Bécancour or Trois-Rivières, Quebec · 200 MW Phase 1, scalable to 1 GW · Hydro-Québec direct-connect
Resilience site (Tier 1)	Gillam, Manitoba · Indigenous co-owned · Manitoba Hydro surplus capture
Nuclear-co-located (Tier 2)	Wesleyville / Port Hope, ON · co-located with OPG SMR / Bruce C development
West-coast latency node	BC Hydro Schedule 1894 site · Pacific-facing fibre to Korea, Japan, Australia
Compute architecture	Korean HBM-anchored accelerator nodes · Canadian-built rack integration · Canadian-staffed operations
Network	Canadian-owned fibre · sovereign DNS · sovereign certificate authority · no third-country root trust

C · GOVERNANCE AND OFFTAKE

Operated by a Crown corporation chartered by Act of Parliament, with a fiduciary mandate to provide sovereign-tier compute to (in order): federal Crown agencies, provincial governments, Indigenous nations under OCAP® / CARE principles, municipalities, Canadian-domiciled regulated entities, and Canadian-domiciled commercial customers. Allied workloads at commercial rates under reciprocal CDSS-compliance. No US-parented entity may host Tier-A workloads under the founding charter.

D · THE MUNICIPAL FLOOR

The application layer reaches down to every Canadian municipality through a published catalogue of CDSS-compliant Canadian software vendors operating on Sovereign Compute North. Fit For Gov is one such application-layer vendor — providing custom municipal websites, citizen portals, council systems, FOIP-released workflows and clerk-of-council tools — but the architecture is vendor-neutral. The point is that a CAO in Annapolis Royal, in Westlock, in Yellowknife, in Saint John, can procure a software stack whose entire jurisdictional chain — vendor, hosting, operations, key custody, support — terminates inside

Canadian law. The substrate that made Hamilton's $18.3 million ransomware recovery a property-tax expense becomes a Crown utility.

THE INTEGRITY CONTRACT

Every record processed by Sovereign Compute North carries a tamper-evident provenance log: residency, custody chain, encryption posture, foreign-process notifications. A Canadian municipality can prove to its citizens that their data has never left Canadian jurisdiction. So can a Crown agency, an Indigenous nation, or an allied sovereign customer.

§ VIII THE CANADIAN DATA SOVEREIGNTY STANDARD · FIVE CLAUSES

What the standard says, in plain language a council can adopt.

The Canadian Data Sovereignty Standard (CDSS) is the regulatory instrument that makes Sovereign Compute North enforceable rather than aspirational. It can be jointly owned by the Communications Security Establishment and its Canadian Centre for Cyber Security, Treasury Board Secretariat, the Office of the Privacy Commissioner of Canada, the Office québécois de la langue française (for Loi 25 alignment), and the First Nations Information Governance Centre (for OCAP® and Indigenous data sovereignty). It can be exported to Korea, Japan, the European Union, the United Kingdom, Australia, New Zealand and India — countries already moving from data-residency to vendor-domicile rules.

01 **Vendor Domicile.**

For Tier-A data — Crown, citizen, Indigenous, health, electoral, judicial, municipal — the controlling parent entity of the cloud, software-as-a-service, or infrastructure vendor must not be subject to the United States CLOUD Act or any equivalent extraterritorial production order regime. Place-of-business in Canada is necessary but not sufficient.

02 **Key Custody.**

Encryption keys protecting Tier-A workloads must be held in hardware-security modules physically located in Canada and operated by entities outside CLOUD-Act-equivalent jurisdiction. Split-key and threshold custody between Crown agencies are required for the highest tier. No vendor escrow.

03 **Operations Residency.**

24/7 site-reliability and administrative operations for Tier-A systems must be staffed by personnel cleared under Canadian law, working from Canadian territory. "Follow-the-sun" support that exits Canadian jurisdiction is prohibited for Tier-A workloads. The human at the console answers to a Canadian court.

04 **Foreign-Process Notification & Audit.**

The vendor is bound to notify the Canadian customer of any foreign-state legal process directed at the customer's data, to the maximum extent permitted by that foreign law, and to provide a quarterly transparency report. Independent Canadian audit rights are conferred on the Office of the Privacy Commissioner and the Canadian Centre for Cyber Security.

05 **Indigenous Data Sovereignty.**

Tier-A data concerning First Nations, Métis or Inuit communities is governed under OCAP® principles (Ownership, Control, Access, Possession) as maintained by the First Nations Information Governance Centre, and under CARE principles (Collective Benefit, Authority to Control, Responsibility, Ethics). Indigenous communities determine their own residency, custody and access regimes; the Crown sovereign cloud accommodates rather than overrides.

WHY THIS IS EXPORTABLE, NOT PAROCHIAL

France, Germany, the European Union, Australia and Korea have each independently arrived at the same conclusion: data-residency rules without vendor-domicile rules do not produce sovereignty. A Canadian standard that names this honestly, that aligns with OCAP® and CARE for Indigenous data, and that builds on ISO/IEC 27001, NIST 800-53, ITSG-33 and Loi 25, can be adopted by every allied country still negotiating its own version. Canada has a one-time opportunity to publish the standard others copy. It will not have that opportunity twice.

§ IX RECOMMENDATIONS · BY AUDIENCE

What each audience can do in the next four quarters.

FOR THE MINISTER OF INNOVATION, SCIENCE AND INDUSTRY & THE MINISTER OF AI AND DIGITAL INNOVATION

1. Publish the Canadian Data Sovereignty Standard (CDSS) in Q3 2026, drafted jointly with CSE/Cyber Centre, TBS, OPC, FNIGC and OQLF, binding Tier-A federal procurement to non-CLOUD-Act-jurisdiction vendors with explicit vendor-domicile, key-custody and operations-residency clauses.
2. Direct the Major Projects Office to designate a flagship Sovereign Compute Campus and run an open RFP under the CDSS for a Korea-led, Canadian-controlled consortium.
3. Use the existing $2 billion Sovereign AI Compute Strategy budget to anchor a first 200 MW phase by FY 2027, with a defined trigger to expand to 1 GW once Korean HBM4/HBM4E offtake is secured and Canadian-domiciled inference-grade silicon partners are identified.

FOR PROVINCIAL PREMIERS · QUEBEC, MANITOBA, BC, ONTARIO, ALBERTA

1. Quebec reserves a tranche of Hydro-Québec's projected 1,000 MW data-centre allocation for a Crown-anchored sovereign compute campus, ringfenced from US-hyperscaler bidding, paired with Loi 25 as the strictest residency-and-domicile jurisdiction in the federation.
2. Manitoba redirects surplus export-bound electrons (~8,000 GWh per year) to a Crown-anchored, Indigenous-co-owned sovereign data-centre corridor.
3. BC reserves a slice of the Schedule 1894 Clean Industry and Innovation cap (1,500 GWh per year) for sovereign-classified workloads.
4. Ontario aligns Pickering B refurbishment and Darlington SMR siting with co-located sovereign compute halls; pairs Bruce C with a Crown-anchored AI training campus.
5. Alberta pairs its December 2024 AI Data Centre Strategy with the CDSS, so that gas-powered AI data centres in Alberta are at least bound to Canadian-jurisdiction vendors.

FOR MUNICIPAL CHIEF ADMINISTRATIVE OFFICERS

1. Run a Hamilton stress test within 90 days: catalogue every workload whose loss or exfiltration would exceed CAD $5 million to recover, and identify which are hosted on CLOUD-Act-jurisdiction vendors.
2. Insist that the federal CDSS include a municipal procurement schedule. Without it, municipalities will continue procuring CLOUD-Act-exposed platforms because Buy Canadian alone does not forbid it.
3. Treat Canadian-domiciled application vendors (Fit For Gov and peers) as the default candidate stack for citizen-facing workflows once a CDSS-compliant Canadian Sovereign Cloud is operational.

FOR SOVEREIGN WEALTH & PENSION COUNTERPARTS · NPS, KIC, KIPE, CPPIB, CDPQ, BCI, ALBERTA HERITAGE, GPIF, FUTURE FUND

1. Joint anchor LP commitment to a CDSS-compliant Sovereign Compute Fund, target CAD $10–20 billion over five years, structured as a Crown-majority infrastructure vehicle.
2. NPS / KIC + CDPQ "Memory–Megawatts" facility: a co-investment vehicle pairing Korean fab capex with Canadian power capex, hedging both sides against Stargate concentration risk and against rising US export-control overhang.

§ X CAVEATS · WHERE THIS PAPER COULD BE WRONG

What would change my mind.

A working paper that does not name its own load-bearing assumptions is not a working paper; it is a press release. The following caveats are the points at which this thesis is most exposed to disconfirmation.

A · THE 13¢/KWH TARIFF IS NOT YET APPROVED

Hydro-Québec's data-centre tariff was filed on 18 February 2026 and is pending Régie de l'énergie approval. Existing customers face a five-year transition. The CAD $0.036/kWh framework rate cited throughout this paper is Hydro-Québec's framework offering to qualifying flexible long-term industrial loads; it is not the published standard industrial tariff and qualification is non-trivial. If the Régie modifies or rejects the filing, the Quebec anchor case weakens.

B · HBM SHARE FIGURES ARE METHODOLOGY-SENSITIVE

Counterpoint Research's HBM share measures shift between bit-shipment and revenue methodologies. The combined Korean position (~75–80%) is robust across methodologies; the SK Hynix vs. Samsung split within that range is not. The "~95% Korean HBM" figure that circulates in some advocacy material is not supported by current data and should not be used.

C · STARGATE EXECUTION RISK IS REAL BUT CUTS BOTH WAYS

The reported staffing and structural difficulties at Stargate (Bloomberg, August 2025; The Information, February 2026) are real and ongoing. This is good news for the Sovereign Compute North case (US compute is harder to build than announced) but does not change the underlying fact that ~7 GW and $400 billion-plus of US compute has been committed and is breaking ground. Do not assume Stargate disappears.

D · "OUTSIDE US DEPENDENCE" IS PARTIAL, NOT TOTAL

Nvidia and AMD GPUs, ASML EUV lithography, and large portions of the cloud-software stack remain US-or-allied-controlled. Sovereign Compute North reduces jurisdictional exposure (CLOUD Act, Section 702 FISA, export-control reach) and power-cost exposure. It does not eliminate technology supply-chain exposure to the United States. The honest framing is "additionally sovereign," not "decoupled."

E · THE CARNIAUX TESTIMONY IS NARROW

Microsoft France's 10 June 2025 admission was on the specific question of whether French data could be guaranteed against US lawful process. Microsoft has stated separately that it litigates and minimises such requests. The legal point — that CLOUD Act jurisdiction reaches data held by US-parent entities regardless of physical location — is uncontested by Microsoft, AWS or Google, and was effectively reiterated by AWS in its July 2025 "Five Facts" response. Treat the testimony as load-bearing for vendor-domicile arguments; do not extend it beyond its actual scope.

F · THE MUNICIPAL SOVEREIGNTY CASE IS NORMATIVE

The City of Hamilton's $18.3 million ransomware recovery is fact. The inference that a Crown-anchored sovereign cloud would have prevented or cheapened that recovery is an argument, not a proof. The evidentiary anchor is that Hamilton's insurer denied coverage on a control deficiency (incomplete MFA deployment) and that the affected workloads ran on commodity infrastructure under foreign jurisdiction. The argument extrapolates from there.

NPSI

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Fit For Gov

A CIVIC-TECHNOLOGY PRACTICE · VICTORIA, BRITISH COLUMBIA

Fit For Gov builds custom municipal websites and digital infrastructure for Canadian municipalities as the modern alternative to vulnerable WordPress installations on foreign hosting. The practice is registered in British Columbia, principal-led, and answers its own telephone. Fit For Gov is the application-layer vendor designed to operate on the Sovereign Compute North substrate proposed in this paper.

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Corrections. The author commits to public correction of any factual error in this document within seventy-two hours of notification. Email or phone, above.

Sovereignty. This document was written under Canadian law, by a Canadian author, on Canadian infrastructure. Every claim is footnoted to a primary source. Every caveat is named. That is the standard the paper is proposing be extended to the compute fabric Canada now needs to build.

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